



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER



Test Report No. 2023 – 0061



Agricultural Machinery – Backhoe GMT BY75P

Test Report Number	2023 – 0061
Date of Issuance	March 3, 2023
Valid Until	March 3, 2028
Agricultural Machinery	Backhoe
Type	Triangular bucket, Three-point hitch-mounted, Hydraulic-powered
Brand	GMT
Model	BY75P
Test Requested by	FIT COREA TRADING PHILS., INC. 10 D. Arellano Street, Brgy. 135, Caloocan City

SUMMARY

Performance Criteria	Manufacturer's Specification	AMTEC Test
Loading capacity, kg/load	280	72.72
Loading rate, kg/h	ND ^a	15359.44
Maximum lift height, mm	2406	2670

^a Not specified by the requesting party.

Note: ND – No Data

OBSERVATIONS

1. Operator's manual	
a. Language	English
b. Safety and warnings	Provided
c. Operating information	Provided
d. Accessories and attachments	Provided
e. Maintenance instruction	Provided
f. Storage	Not provided
g. Handling, reception, transportation, assembly, and installation	Not provided
h. Specifications	Provided
i. Dismantling and disposal	Dismantling only
j. Warranty	Provided
k. Parts list	Provided
2. Previous test reports of similar brand and model	None
3. Manufacturer	GMT CO., LTD. 1431-10, Nonsanpyeongya-ro, Noseong-myeon, Nonsan, Chungcheongnam-do, Korea
4. Serial number	Y75PS0003
5. Four-wheel tractor used	40.5 kW TYM 5825R
6. Number of operators	One
7. Cleaning of parts	Parts are accessible for cleaning.
8. Adjusting and repairing of parts	Parts are accessible for repairs.
9. Transporting of the machine	The backhoe is mounted to a four-wheel tractor. The hydraulic outrigger is lifted and the arm and bucket are folded using hydraulic controls prior to transport.
10. Handling and stability when the machine is working and turning	The machine operation is stationary and is stabilized with a hydraulic outrigger.
11. Safety features	Warning stickers are provided.
12. Failure or abnormalities that may be observed on the machine or its component parts during and after the operation	None
13. Other remarks	a. The backhoe is operated while fixed in place using controls that are within reach of the operator. b. The backhoe arm may be moved laterally using hydraulic controls. c. The backhoe was tested for soil transport. Digging function was not tested.

PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing / Commercial
Date of Test	February 9, 2023
Location of Test	Brgy. Unang Sigaw, Quezon City 14°39'39.60" N, 120°59'56.40" E
Items Tested	Operator's Manual, Noise Level, Fuel Consumption, Loading Capacity, Loading, Rate, Maximum Lift Height, and Number of Operators

METHODS OF TEST

The backhoe was tested based on the AMTEC Methods of Test.

DESCRIPTION OF THE MACHINE

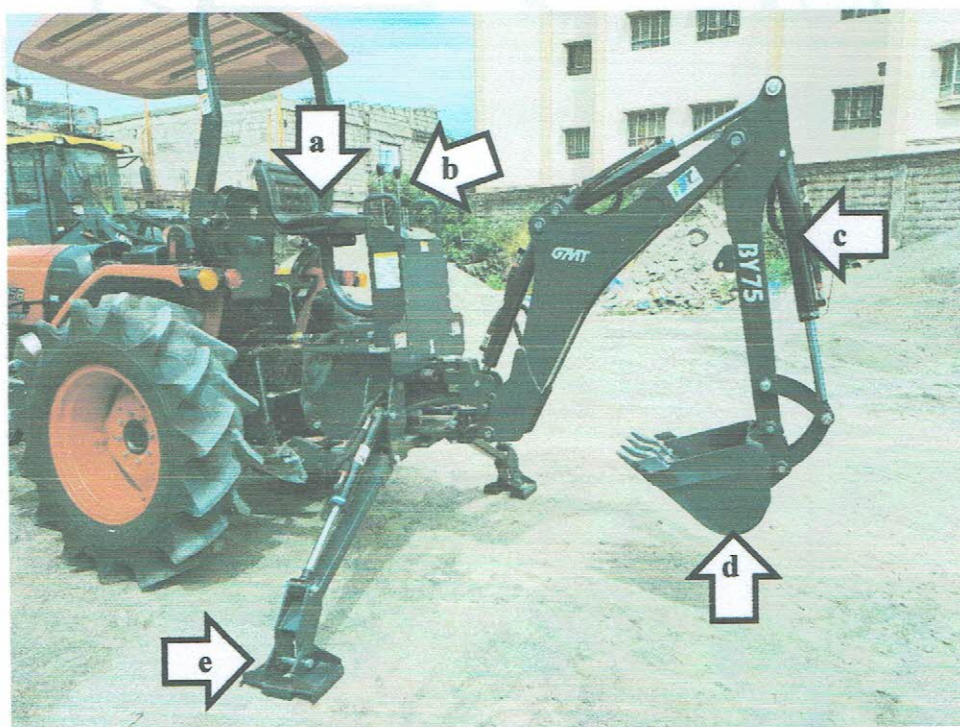


Figure 1. Parts of the GMT BY75P Backhoe powered by 40.5 kW TYM 5825R Four-Wheel Tractor include the following: (a) Operator's seat, (b) Controls, (c) Backhoe arm, (d) Bucket, and (e) Stabilizer.

SPECIFICATIONS

Item	Manufacturer's Specification ^a	AMTEC Verification
1 Main structure		
1.1 Overall dimensions, transport mode, mm		
1.1.1 Length	1189	2510
1.1.2 Width	1737/2256	1650
1.1.3 Height	2284	2199
1.2 Weight, kg	608	NM ^b
2 Total arm length, mm	4050	3100
3 Bucket		
3.1 Dimensions, L × W × H, mm	560 × 340 × 320	440 × 480 × 290
3.2 Opening dimension, L × W, mm	420 × 330	400 × 395
3.3 Shape	Square	Triangular
3.4 Capacity, m ³	0.03	0.054 ^c
3.5 Bucket angle, °		
3.5.1 Maximum dump angle	92	12.8 ^d
3.5.2 Maximum rollback angle	57	27.4 ^d
3.6 Reach, mm		
3.6.1 Maximum reach of bucket, mm	3140	2510
3.6.2 Maximum reach of bucket at maximum lift height, mm	2406	1470
3.6.3 Maximum reach of bucket at collecting position, mm	2286	2435
3.7 Ground clearance at transportation mode, mm	410	590
3.8 Material	Steel structure 400	ND

^a The data were obtained from the specifications sheet provided by the requesting party.

^b Weight was not verified as there was no available scale on site.

^c Measured based on loading capacity during the test.

^d Measured from the horizontal axis.

Note: ND – No Data, NM – Not Measured

SPECIFICATIONS

Item		Manufacturer's Specification ^a	AMTEC Verification
4	Tractor used		
4.1	Brand	TYM	TYM
4.2	Model	5825R	5825R
4.3	Serial number	ND	ND
4.4	Make	Korea	Korea
4.5	Rated power, kW	40.5	40.5 ^a
4.6	Rated speed, rpm	2600	2600 ^a
4.7	Displacement, cm ³	2286	2286 ^a
4.8	Fuel type	Diesel	Diesel
4.9	Cooling system	Water-cooled	Water-cooled
4.10	Starting system	ND	Electric
5	Safety features	Boom and swing fixtures	None
6	Special features	None	None

^a The data were obtained from the specifications sheet provided by the requesting party.

Note: ND – No Data, NM – Not Measured

RESULTS

Item		Data
1	Test site condition	
1.1	Location of test field	Brgy. Unang Sigaw, Quezon City 14°39'39.60" N, 120°59'56.40" E
1.2	Soil condition	
1.2.1	Material used	Sand
1.2.2	Bulk density, kg/m ³	1347.12
1.2.3	Moisture content, % _{db}	3.93
2	Performance test	
2.1	Tractor gear shift setting	1st gear - Medium
2.2	Loading capacity, kg/load	72.72
2.3	Loading rate, kg/h	15359.44
2.4	Fuel consumption, L/h	3.45
2.5	Noise at operator's ear level, dB(A)	83.5

OBSERVATIONS

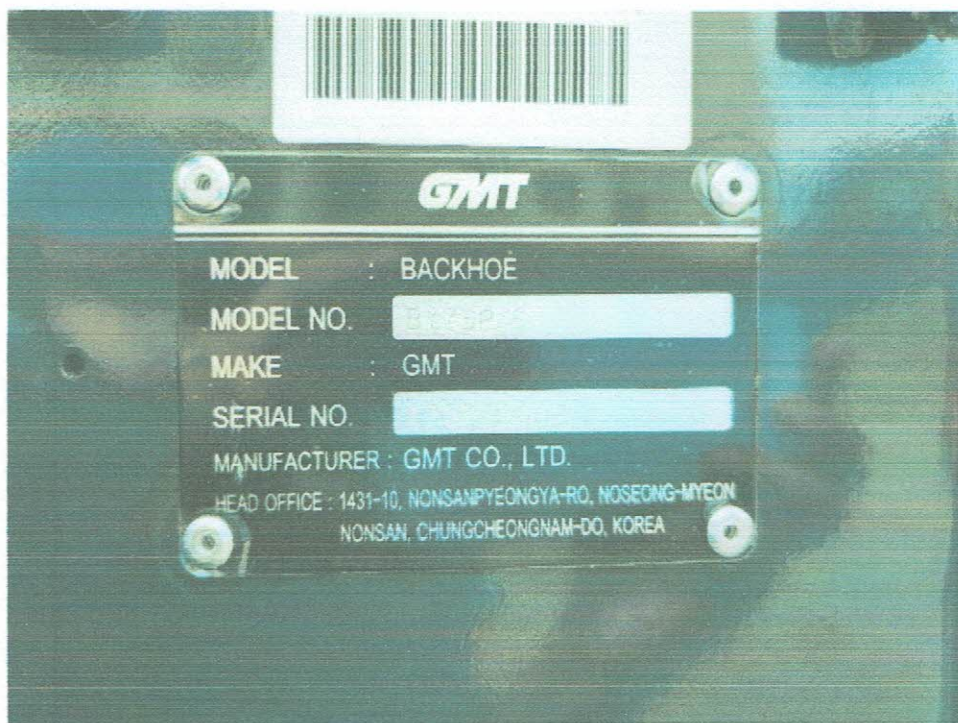


Figure 2. Nameplate of the GMT BY75P Backhoe.



Figure 3. Markings and labeling of the GMT BY75P Backhoe.

OBSERVATIONS (Cont'n)

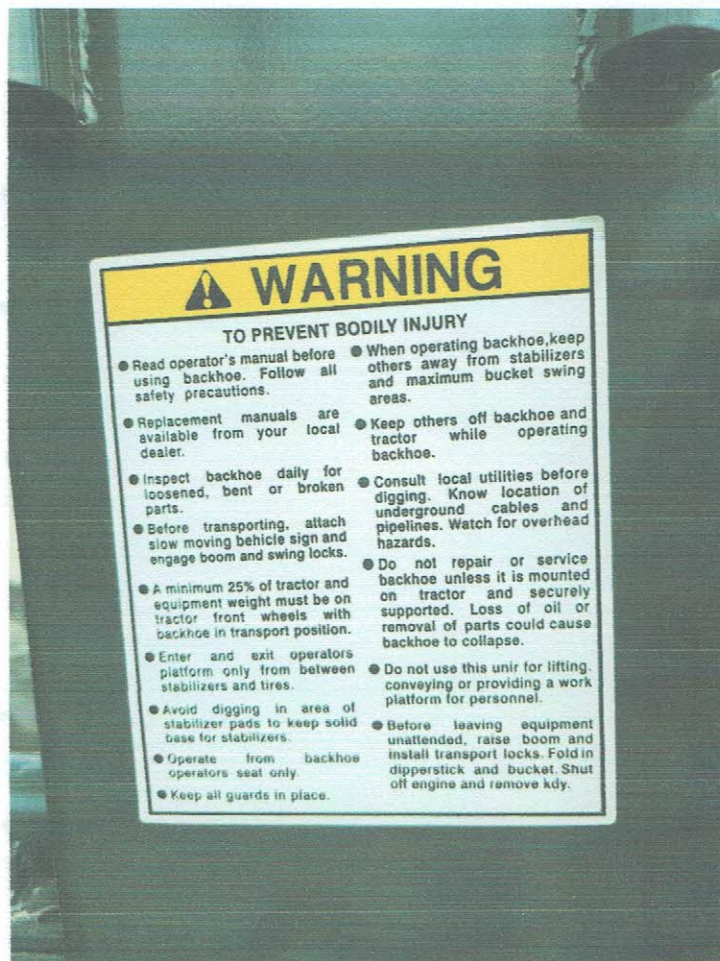


Figure 5. Warning sticker of the GMT BY75P Backhoe.

OBSERVATIONS (Cont'n)



Figure 4. Warning stickers, markings, and labeling of the GMT BY75P Backhoe.

Tested by:


KAREN CYBILL F. CABANGLAN

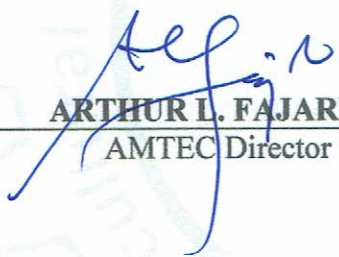
Test Engineer

Verified by:


MARIE JEHOA B. REYES

Test Engineer

Approved for Release:


ARTHUR L. FAJARDO

AMTEC Director

MAR 06 2023

Date

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